

Inside Indecomposability

Some Old, Some New and Some Future Results on the Classification of Indecomposable Cycle Sets

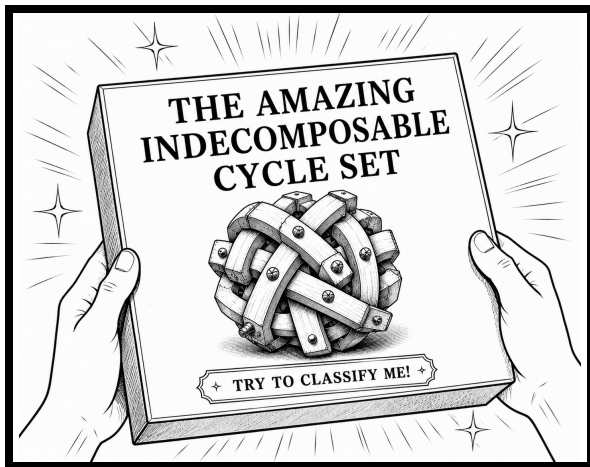
Carsten Dietzel



Braiding new ground:
Vertex algebras meet the Yang-Baxter Equation

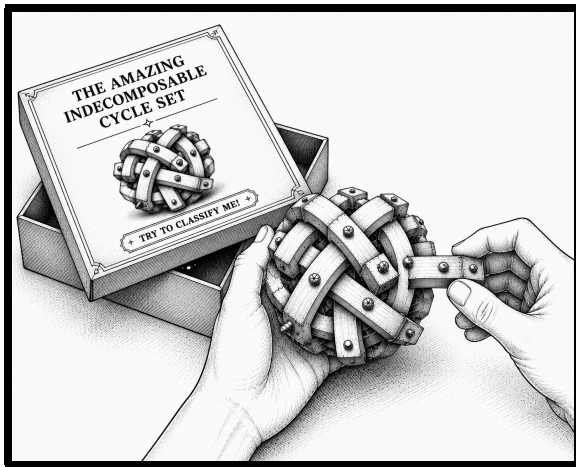
Maynooth, 18th September 2026

What's Inside?



- ➊ Getting in touch with cycle sets
- ➋ Indecomposable cycle sets of size p^2
- ➌ Extensions of indecomposable cycle sets
- ➍ Computing indecomposable cycle sets of small size

Getting in touch with cycle sets



What is a cycle set?

Definition: Cycle set

A **cycle set** is a set X with a binary operation $* : X \times X \rightarrow X$ such that

- for all $x \in X$, the map $\sigma_x : X \rightarrow X; y \mapsto x * y$ is bijective,
- the **cycloid equation**

$$(x * y) * (x * z) = (y * x) * (y * z)$$

is satisfied for all $x, y, z \in X$.

If the **diagonal map** $T_X : X \rightarrow X; x \mapsto x * x$ is bijective, X is **nondegenerate**.

- **Convention:** Throughout the talk, cycle sets are finite and nondegenerate.
- **Why cycle sets?** If X is nondegenerate cycle set, then there is a well-defined map

$$r : X \times X \rightarrow X \times X; \quad (x * y, x) \mapsto (y * x, y).$$

This map satisfies $r^2 = \text{id}_{X \times X}$ and the **Yang-Baxter equation**

$$r_{12} \circ r_{23} \circ r_{12} = r_{23} \circ r_{12} \circ r_{23}$$

Where $r_{12} = r \times \text{id}_X$ and $r_{23} = \text{id}_X \times r$.

The permutation group of a cycle set

Definition: Permutation group

Let X be a cycle set, then its **permutation group** is the group

$$\mathcal{G}(X) = \langle \sigma_x : x \in X \rangle \leq \text{Sym}_X.$$

Conventions: $\lambda_x = \sigma_x^{-1}$. Furthermore, λ_g denotes the action of an element $g \in \mathcal{G}(X)$ on X .

Definition: Indecomposability

A cycle set X is **indecomposable** if for each decomposition $X = A \sqcup B$ into sub-cycle sets A, B , we have either $A = \emptyset$ or $B = \emptyset$.

Theorem [Etingof, Schedler, Soloviev (1999)]

A cycle set X is indecomposable if and only if $\mathcal{G}(X)$ acts transitively on X .

Examples

Example 1: Cyclic cycle sets

$X = \mathbb{Z}_n$ is a cycle set under the operation $x * y = y + 1$.
 $\mathcal{G}(X) \cong \mathbb{Z}_n$.

Theorem [Etingof, Schedler, Soloviev (1995)]

Let X be an indecomposable cycle set of prime size p , then $X \cong \mathbb{Z}_p$ with $x * y = y + 1$.

Example 2: A cycle set of size 4

$X = \{1, 2, 3, 4\}$ can be turned into a cycle set with the multiplication table

*	1	2	3	4
1	2	1	3	4
2	4	3	1	2
3	1	2	4	3
4	3	4	2	1

$\mathcal{G}(X) \cong \mathbb{D}_8$.

The retraction of a cycle set

Observations

- In general, the canonical map $\sigma : X \rightarrow \mathcal{G}(X); \quad x \mapsto \sigma_x$ is **not injective**.
- For $x \in X$, let $[x] = \sigma_x$. Then its **retraction** $X_{\text{ret}} = \sigma(X)$ is a cycle set under the (well-defined) operation $[x] \cdot [y] = [x \cdot y]$.
- X_{ret} might again retract nontrivially!

Retractable / Irretractable

A cycle set X is **retractable** if $|X_{\text{ret}}| < |X|$. Else X it is **irretractable**.

⇒ Irretractability means that X does not have repeated permutations.

Multipermutation

For a cycle set X , let $X^{(0)} = X$ and $X^{(i+1)} = X_{\text{ret}}^{(i)}$.

⇒ The **multipermutation level** of a cycle set X is the smallest integer n with $|X^{(n)}| = 1$ if existent, and ∞ else.

Indecomposable cycle sets of size p^2



Hall- π -subgroups

Let $n = \prod_p p^{v_p}$ and π a set of primes. Define

$$n_\pi = \prod_{p \in \pi} p^{v_p}; \quad n_{\pi'} = \prod_{p \notin \pi} p^{v_p}.$$

Definition: Hall- π subgroup

- Given a set of prime numbers π , call a finite group G a **π -group** if $|G|_\pi = |G|$.
- A subgroup $H \leq G$ is called a **Hall- π subgroup** if $|H| = |G|_\pi$

Fact

Let G be finite. Then

G is solvable.



G has Hall- π -subgroups for all π .

Theorem [Etingof, Schedler, Soloviev (1995)]

$\mathcal{G}(X)$ is solvable for each cycle set X .

A group-theoretic lemma

Lemma

Let G be a finite group and let $A, B \leq G$ be subgroups such that $\gcd((G:A), (G:B)) = 1$, then $AB = G$.

Proof.

- $(G:A \cap B)$ is divisible by $(G:A)$, $(G:B) \Rightarrow (G:A \cap B) \geq (G:A) \cdot (G:B)$.

$$\Rightarrow \Rightarrow |AB| = \frac{|A| \cdot |B|}{|A \cap B|} = \frac{|G| \cdot (G:A \cap B)}{(G:A) \cdot (G:B)} \geq |G|.$$

$$\Rightarrow AB = G.$$

A corollary

Let X be finite and $G \leq \mathfrak{S}_X$ transitive. Let $\pi_X = \{p : p \mid |X|\}$.

- Let X be finite and $G \leq \mathfrak{S}_X$ transitive.
- Let $\pi_X = \{p : p \mid |X|\}$.
- If H is a Hall- π -subgroup of G , then H is transitive.

Proof. $(G:G_x) = |X|$ is coprime to $(G:H) \Rightarrow HG_x = G \Rightarrow H$ is transitive.

The Feingesicht construction

- Let π, π' be disjoint sets of primes
- Let $(X, *_{\pi}), (X, *_{\pi'})$ be two cycle sets on a set X where $\mathcal{G}(X, *_{\pi}), \mathcal{G}(X, *_{\pi'})$ is a π - resp. π' -group.
- Assume that they satisfy the **mixed condition**

$$(x *_{\pi} y) *_{\pi'} (x *_{\pi} z) = (y *_{\pi'} x) *_{\pi} (y *_{\pi'} z),$$

then $x * y = (x *_{\pi'} x) *_{\pi} (x *_{\pi'} y)$ is the operation of a cycle set on X .

- $\mathcal{G}(X, *)$ has $\mathcal{G}(X, *_{\pi})$ and $\mathcal{G}(X, *_{\pi'})$ as Hall- π resp. Hall- π' -subgroups.
- Given π, π' , every cycle set with $\mathcal{G}(X)$ a $(\pi \cup \pi')$ -group can be decomposed into π - resp. π' -components.

General strategy

The goal

Let p be a prime. Determine, up to isomorphism, all indecomposable cycle sets of size p^2 .

Definition: A cycle set X is of **p -type** if X is indecomposable and $\mathcal{G}(X)$ is a p -group.

Strategy

- Determine indecomposable cycle sets of **p -type**.
- Use the method of Feingesicht to extend the cycle sets of p -type to get all indecomposable cycle sets of size p^2 .

Generalization: X is of **π -type**, if

- X is indecomposable,
- $p \mid |X| \Leftrightarrow p \mid |\mathcal{G}(X)|$ for all primes p .

X of p -type, $\text{mpl}(X) = 1$

We will always assume that $|X| = p^2$.

Multipermutation level 1

If X is of multipermutation level 1, then $x * y = \sigma(y)$ for some $\sigma \in \text{Sym}(X)$.
 $\mathcal{G}(X) = \langle \pi \rangle$ has to act transitively on $X \Rightarrow \pi$ is a p^2 -cycle.
 $\Rightarrow$ we can choose $X = \mathbb{Z}_{p^2}$ and $x * y = y + 1$.

Obtained cycle sets

$X = (\mathbb{Z}_{p^2}, *)$, $x * y = y + 1$.

This was easy! The case of multipermutation level 2 is slightly more involved.

Parametrizing cycle sets of p -type

Some group theory

- The p -Sylow subgroups of $\text{Sym}(\{1, \dots, p^2\})$ are isomorphic to $\mathbb{Z}_p \wr \mathbb{Z}_p = \mathbb{Z}_p \ltimes \mathbb{Z}_p^p$.
- Identifying $\{1, \dots, p^2\}$ with $\mathbb{Z}_p \times \mathbb{Z}_p$, the elements of $\mathbb{Z}_p \wr \mathbb{Z}_p$ are given by

$$\pi(x, a) = (x + \Delta, a + \Phi(x))$$

with $\Delta \in \mathbb{Z}_p$ and some mapping $\Phi : \mathbb{Z}_p \rightarrow \mathbb{Z}_p$.

Consequence

- A cycle set X of p -type with $|X| = p^2$ can be parametrized as $X = \mathbb{Z}_p \times \mathbb{Z}_p$ with

$$(x, a) * (y, b) = (y + \Delta(x, a), b + \Gamma(x, a, y)).$$

- However, determining the functional parameters Δ, Φ is almost impossible to do by hand.

X of p -type, $\text{mpl}(X) = 2$

Procedure

Let X be a p -type cycle set of multipermutation level 2, $|X| = p^2$.

- X_{ret} is indecomposable of size $p \Rightarrow X_{\text{ret}} \cong \mathbb{Z}_p$ with $a * b = b + 1$.
- Therefore $(x, a) * (y, b) = (y + 1, b + \Gamma(x, y))$, where

$$\Gamma(x, z) + \Gamma(y + 1, z + 1) = \Gamma(y, z) + \Gamma(x + 1, z + 1).$$

$\Rightarrow \Gamma(x, y) = \Delta(y) + \Phi(y - x)$ with $\Delta, \Phi : \mathbb{Z}_p \rightarrow \mathbb{Z}_p$ and $\Phi(0) = 0$.

- Demanding $\Phi \neq 0$ ensures indecomposability.

Obtained cycle sets

$$X = \mathbb{Z}_p \times \mathbb{Z}_p$$

$$(x, a) * (y, b) = (y + 1, b + S \cdot \Delta(y) + \Phi(y - x)),$$

where $\Phi : \mathbb{Z}_p \rightarrow \mathbb{Z}_p$, $S \in \mathbb{Z}_p$, and

$$\Delta(x) = \begin{cases} 1 & x = 0 \\ 0 & x \neq 0. \end{cases}$$

X of p -type, $\text{mpl}(X) = \infty$

Definition: $\sigma(X) = \{\sigma_x : x \in X\}$.

The strategy

- Let F be the fix with respect to the brace structure on $\mathcal{G}(X)$.
- Then for all $x \in \mathbb{Z}_p$, the set $\{\sigma_{(x,a)} : x \in \mathbb{Z}_p\} \subseteq \mathcal{G}(X)$ is a left coset of F .
- Using the λ -action of F , one obtains $\sigma(X) = F \circ \sigma_{x_0} \circ F$.

Obtained cycle sets

$$X = \mathbb{Z}_p \times \mathbb{Z}_p$$

$$(x, a) * (y, b) = (y + a, b + \Phi(y - x)),$$

for some $\Phi : \mathbb{Z}_p \rightarrow \mathbb{Z}_p$ with $\Phi(z) = \Phi(-z)$ for all z .

From cycle sets of p -type to all cycle sets

Procedure

- $|\mathcal{G}(X, *)_{p'}| > 1$ only occurs if $(X, *)_p$ is irretractable.
- If $(X, *)_p$ is irretractable, then $(x, a) *_{p'} (y, b) = \gamma(y, b)$ for some $\gamma \in \text{Aut}(X, *)_p$.
- We can ignore β . Applying the Feingesicht method, we get

$$\begin{aligned}(x, a) * (y, b) &= ((x, a) *_{\rho} (x, a)) *_{\rho'} ((x, a) *_{\rho} (y, b)) \\ &= \gamma((x, a) *_{\rho} (y, b))\end{aligned}$$

- Automorphisms of $(X, *)_p$ are of the form $(x, a) \mapsto (\alpha x + \beta, \alpha a)$ ($\alpha \in \mathbb{Z}_p^*$, $\beta \in \mathbb{Z}_p$) where $\Phi(\alpha z) = \alpha(\Phi(z))$.

Remaining cycle sets

$$\begin{aligned}X &= \mathbb{Z}_p \times \mathbb{Z}_p \\ (x, a) * (y, b) &= (\alpha(y + a), \alpha(b + \Phi(y - x))),\end{aligned}$$

where $\Phi(z) = \Phi(-z)$, $\alpha \neq 0$ and $\Phi(\alpha z) = \alpha(\Phi(z))$ for all z .

Theorem [D., Properzi, Trappeniers (2024)]

Let p be a prime. Then each indecomposable cycle set is isomorphic to one of the following

- $X = \mathbb{Z}_p \times \mathbb{Z}_p$,

$$(x, a) * (y, b) = (y + 1, b + S \cdot \Delta(y) + \Phi(y - x)),$$

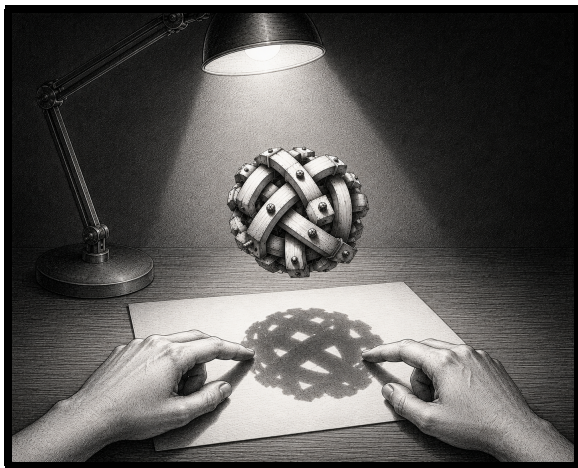
with $S \in \mathbb{Z}_p$, $\Phi : \mathbb{Z}_p \rightarrow \mathbb{Z}_p$, satisfying $(S, \Phi) \neq (0, 0)$.

- $X = \mathbb{Z}_p \times \mathbb{Z}_p$,

$$(x, a) * (y, b) = (\alpha(y + a), \alpha(b + \Phi(y - x))),$$

with $\Phi : \mathbb{Z}_p \rightarrow \mathbb{Z}_p$, $\alpha \neq 0$, satisfying $\Phi(z) = \Phi(-z)$ and $\Phi(\alpha z) = \alpha(\Phi(z))$

Extensions of indecomposable cycle sets



The Cedó-Okniński theorem

Theorem [Cedó, Okniński (2023)]

Let X be an indecomposable cycle set of squarefree size $|X| = p_1 p_2 \dots p_k$, then

- X is retractable.
- If a prime p divides $|\mathcal{G}(X)|$, then $p = p_i$ for some $1 \leq i \leq k$.
- $\text{mpl}(X) \leq k$.

Fact If X is an indecomposable cycle set and $Y = X_{\text{ret}}$, then Y is indecomposable and $|Y|$ divides $|X|$.

⇒ We can use the Cedó-Okniński theorem to recursively construct indecomposable cycle sets of squarefree size!

Our strategy

- Find a method to construct an indecomposable cycle set from its retraction.
- Recursively apply it to cycle sets of squarefree size.

X-graded modules

Definition: X-graded G-module

Let G be a group, acting on a set X via $(g, x) \mapsto g \cdot x$.

An **X-graded G-module** is an abelian group A with a decomposition into graded components

$$A = \bigoplus_{x \in X} A_x$$

and an action $G \times A \rightarrow A$; $(g, a) \mapsto g \cdot a$ such that:

- for all $g \in G$, the map $a \mapsto g \cdot a$ is an automorphism of A ,
- for all $g \in G, x \in X$: $g \cdot A_x = A_{g \cdot x}$.

Definition: Scattering

Let A be an X -graded G -module. Then its **scattering** is the set

$$A^{\text{sc}} = \{(x, a) \in X \times A : a \in A_x\},$$

together with the G -action $g \cdot (x, a) = (g \cdot x, g \cdot a)$.

Twisted extensions of cycle sets by graded modules

Definition: Constant extension

A cycle set homomorphism $f : Y \rightarrow X$ is called a (constant) extension if

- f is surjective,
 - for all $x, y \in X$: $f(x) = f(y) \Rightarrow \sigma_x = \sigma_y$.
-
- All extensions are assumed to be constant.
 - Universal prototype of an extension: $\text{ret} : X \rightarrow X_{\text{ret}}$.

Definition: A-extension

Let X be a cycle set and let A be an X -graded $\mathcal{G}(X)$ -module. An extension of cycle sets $f : Y \rightarrow X$ is a twisted extension of X by A if Y is equipped with a group action $A \times Y \rightarrow Y$; $(a, y) \mapsto a + y$ such that

- for all $x \in X$, A_x acts regularly on $f^{-1}(x)$,
- if $x \neq y$, A_y acts trivially on $f^{-1}(x)$,
- for all $x, y \in Y$, $a \in A$ we have: $x * (a + y) = (\sigma_x \cdot a) + x * y$.

$\Rightarrow$ for the $\mathcal{G}(X)$ -module B^X , B an abelian group, one recovers a definition by Lebed and Vendramin!

Construction of twisted extensions

Construction: $X \otimes_{\Phi} A$

- Let X be a cycle set and A an X -graded $\mathcal{G}(X)$ -module.
- Let $\Phi : X \times X \rightarrow A$ be a **twisted 2-cocycle**, that is, a map satisfying the following conditions:

$$\Phi(x, y) \in A_y \quad (x, y \in X),$$

$$\Phi(x, z) + \lambda_x \cdot \Phi(x * y, x * z) = \Phi(y, z) + \lambda_y \cdot \Phi(y * x, y * z).$$

- Construct $X \otimes_{\Phi} A = A^{\text{sc}}$ with the cycle set operation

$$(x, a) * (y, b) = \sigma_x \cdot (y, b + \Phi(x, y)) = (x * y, \sigma_x \cdot (b + \Phi(x, y))),$$

together with the projection

$$p : X \otimes_{\Phi} A \twoheadrightarrow X; \quad (x, a) \mapsto x.$$

⇒ All twisted extensions of X by A can be recovered in that way!

Equivariant maps

We want to focus on twisted extensions defined by equivariant maps:

Definition: Equivariant map

Let X be a cycle set and A an X -graded $\mathcal{G}(X)$ -module. Then a map $\Phi : X \times X \rightarrow A$ is **equivariant** if

$$\begin{aligned}\Phi(x, y) &\in A_y & \forall x, y \in X \\ \Phi(\lambda_g(x), \lambda_g(y)) &= g \cdot \Phi(x, y) & \forall g \in \mathcal{G}(X); x, y \in X.\end{aligned}$$

Equivariant maps are twisted 2-cocycles:

$$\Phi(x, z) + \lambda_x \cdot \Phi(x * y, x * z) = \Phi(x, z) + \Phi(y, z) = \Phi(y, z) + \lambda_y \cdot \Phi(y * x, y * z).$$

Coprime extensions

In the following, we focus on indecomposable cycle sets.

Definition: Coprime extension

Let $f : Y \rightarrow X$ be an extension of indecomposable cycle sets. Then f is **coprime** if

$$\gcd\left(|\mathcal{G}(X)|, \frac{|Y|}{|X|}\right) = 1.$$

Coprimality allows a parametrization in terms of equivariant maps:

Theorem [D. (2024)]

Let $f : Y \rightarrow X$ be a coprime extension of indecomposable cycle sets. Then:

- f is equivalent to $p : X \otimes_{\Phi} A \rightarrow X$ for some equivariant map $\Phi : X \times X \rightarrow A$ to some X -graded $\mathcal{G}(X)$ -module A .
- If $(A, \Phi), (B, \Gamma)$ are two such parametrizations, then there is an isomorphism $\alpha : A \rightarrow B$ such that $\Gamma = \alpha \circ \Phi$.

Indecomposable cycle sets of size p and pq

Theorem [Etingof, Schedler, Soloviev (1995)]

Let X be an indecomposable cycle set of prime size p , then $X \cong \mathbb{Z}_p$ with $x * y = y + 1$.

Theorem [Jedlička, Pilitowska (2023)]

Let X be an indecomposable cycle set of size pq such that $|X^{(1)}| = p$, then $X \cong \mathbb{Z}_p \times \mathbb{Z}_q$ with

$$(x, a) * (y, b) = (y + 1, b + \Gamma^0(y - x)),$$

where $\Gamma^0 : \mathbb{Z}_p \rightarrow \mathbb{Z}_q$ is nonconstant.

Outline: Cycle sets of size pqr

We only discuss the case $\text{mpl}(X) = 3$ where $|X| = pqr$, $|X^{(1)}| = pq$, $|X^{(2)}| = p$.

Strategy

- We know that $X^{(1)} = \mathbb{Z}_p \times \mathbb{Z}_q$ with $(x, a) * (y, b) = (y + 1, b + \Gamma^0(y - x))$ for some nonconstant $\Gamma^0 : \mathbb{Z}_p \rightarrow \mathbb{Z}_q$.
- Find all $X^{(1)}$ -graded $\mathcal{G}(X^{(1)})$ -modules A with $A_{(x_0, a_0)} \cong \mathbb{Z}_r$.
- This is (mostly) equivalent to finding H -module structures on $\mathbb{Z}_r$, where $H = \mathcal{G}(X^{(1)})_{(x_0, a_0)}$.
- For the determined $\mathcal{G}(X^{(1)})$ -modules A , find all equivariant parameter functions Φ .

Main challenge: understanding the orbit structure of the stabilizing group H .

Cycle sets of size pqr and multipermutation level 3, Part I

Case I: $\Gamma^0(x) = \alpha \cdot \xi^x \Rightarrow \mathcal{G}(X^{(1)})$ acts regularly on $X^{(1)} \Rightarrow A \cong A_{(0,0)}^{X^{(1)}}$.

Theorem [D. (2024)]

Let X be an indecomposable cycle set of size pqr with $X^{(1)} = \mathbb{Z}_p \times \mathbb{Z}_q$ where

$$(x, a) * (y, b) = (y + 1, b + \xi^{y-x}).$$

for some $1 \neq \xi \in \mathbb{Z}_q$ with $\xi^p = 1$.

Then there is a map $\Phi^0 : X^{(1)} \rightarrow \mathbb{Z}_r$ such that $X = \mathbb{Z}_p \times \mathbb{Z}_q \times \mathbb{Z}_r$ with

$$(x, a, s) * (y, b, t) = (y + 1, b + \xi^{y-x}, t + \Phi^0(y - x, b - \xi^{y-x}a)).$$

Main idea: first determine $\mathcal{G}(X^{(1)})_{(0,0)}$ -equivariant functions $\Phi^0((0, 0), (-, -))$, then shift $(0, 0)$ around.

Case II: $A \cong A_{(0,0)}^{\chi(1)}$; Γ^0 is not of the form $\Gamma^0(x) = \alpha \cdot \xi^x$.

Construction A

- Let $\Phi_1^0 : \mathbb{Z}_q \rightarrow \mathbb{Z}_r$ and $\Phi_2^0 : \mathbb{Z}_p^* \rightarrow \mathbb{Z}_r$ be given.
- Construct the cycle set $X = \mathbb{Z}_p \times \mathbb{Z}_q \times \mathbb{Z}_r$ with

$$(x, a, s) * (y, b, t) = \begin{cases} (y+1, b + \Gamma^0(y-x), t + \Phi_1^0(b-a)) & x = y \\ (y+1, b + \Gamma^0(y-x), t + \Phi_2^0(y-x)) & x \neq y. \end{cases}$$

- This results in a cycle set of multipermutation level ≤ 3 .
- This construction can easily be generalized!

Construction B

- Let $K = \langle \Gamma_x \rangle \leq \mathbb{Z}_q^{\mathbb{Z}_p}$ where $\Gamma_x(y) = \Gamma^0(y - x)$.
- Let $\chi : \mathbb{Z}_q^{\mathbb{Z}_p} \rightarrow \mathbb{Z}_r^*$; $f \mapsto \prod_{x \in \mathbb{Z}_p^*} \xi_x^{f(x)}$ for $\xi_x \in \mathbb{Z}_q$ ($x \in \mathbb{Z}_p^*$) with $\xi_x^p = 1$.
- Put $\chi_x(f) = \prod_{y \in \mathbb{Z}_p^*} \xi_y^{f(x+y)}$ and suppose that

$$0 \notin \mathcal{N}_\chi = \{x \in \mathbb{Z}_p : \forall f \in K : f(x) = f(0) = 0 \Rightarrow \chi_x(f) = 1\}.$$

- define a map $\psi_x^0 : \mathbb{Z}_p \times \mathbb{Z}_q \rightarrow \mathbb{Z}_r$ by:

$$\psi_x^0(y, f(x)) = \begin{cases} 0 & x \neq y, \\ \chi_x(f) & x = y. \end{cases} \quad \text{for } f \in K \text{ with } f(0) = 0,$$

and put $\Phi^0 = \sum_{x \in \mathcal{N}_\chi} c_x \psi_x^0$ (not all $c_x \in \mathbb{Z}_r$ zero).

- suppose that $\Gamma^0(-x_0) = 1$ and construct $X = \mathbb{Z}_p \times \mathbb{Z}_q \times \mathbb{Z}_r$ with

$$\begin{aligned} & (x, a, s) * (y, b, t) \\ &= \\ & (y + 1, b + \Gamma_x(y), \chi_y(\Gamma_x) \cdot (t + \chi_{x-y}(a \cdot \Gamma_{x_0}) \cdot \Phi^0(x - y, b - a \cdot \Gamma_{x_0}(y - x)))) \end{aligned}$$

Theorem [D. (2024)]

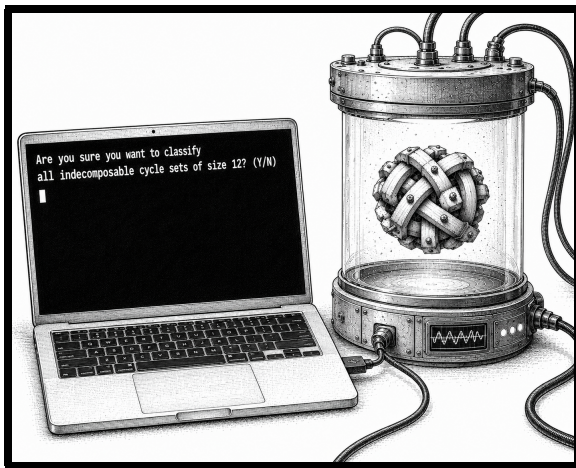
Let X be an indecomposable cycle set of size pqr such that $X^{(1)} = \mathbb{Z}_p \times \mathbb{Z}_q$ with the cycle set operation

$$(x, a) * (y, b) = (y + 1, b + \Gamma^0(y - x))$$

where $\Gamma^0(x)$ is *not* of the form $\alpha \cdot \xi^x$. Then X is isomorphic to a cycle set obtainable by Construction A or Construction B.

- The complexity of Construction B and the methods used to obtain the classification suggest that it is unlikely that an explicit description of all indecomposable cycle sets of size pqr s can be given!

Computing indecomposable cycle sets of small size



Some computational results

Theorem [Akgün, Mereb, Vendramin (2020)]

The function $C(n)$ which enumerates nonisomorphic cycle sets of given size n , has the following values for $n \leq 10$:

n	1	2	3	4	5	6	7	8	9	10
$C(n)$	1	2	5	23	88	595	3 456	34 530	321 931	4 895 272

Theorem [Boegarts, Van Caudenberg, Vendramin (2025)]

$C(11) = 77\,182\,093$.

Strategy

- Split the search space by cycle structure of diagonal $T(x) = x * x$.
- Use constraint programming to find cycle sets up to isomorphism.
- Use lexicographical symmetry breaking to avoid isomorphism checking!

A computational problem

Problem

Find ways to compute indecomposable cycle sets of a given size!

This is more difficult than expected!

Obstacles

- We know all cycle sets of size 8 $\Rightarrow$ only indecomposable cycle sets of size 8 are fully known, without a general structure theorem.
- Computing all cycle sets of size n , and filtering for indecomposables, is incredibly wasteful (there are 77 182 093 cycle sets of size 11, but only one is indecomposable).

The idea

- Indecomposable cycle sets of non-prime size are imprimitive.
- So there is a block structure to work with!

New indecomposable cycle sets of small size

Work in Progress with: Andrew Darlington, Silvia Properzi and Magdalena Wiertel

Definition: $k(X)$

Let X be an indecomposable cycle.

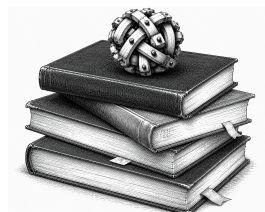
- For a prime p , let M_p be the maximal normal p -subgroup of $\mathcal{G}(X)$.
- Define $k(X)$ as the size of orbits of M_p where p is the least prime with M_p .

Strategy

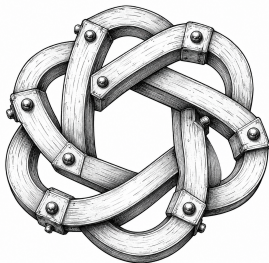
- $\mathcal{G}(X)$ is a transitive solvable group on 12 elements.
 - For $p \mid |\mathcal{G}(X)|$, let $M_p \leq \mathcal{G}(X)$ be the maximal normal p -subgroup.
 - Split the search space by: $k(X)$, diagonal map $T(x) = x * x$.
 - Determine cycle sets by constraint programming.
 - Remove cycle sets with $k(X) \neq k$.
-
- Not every diagonal is compatible with a prescribed block system $\Rightarrow$ significant reduction of cases!

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Thanks for your attention!

